Constraining Primordial Non-Gaussianity with Multi-Tracer Line Intensity Mapping: A Realistic SPHEREx Forecast

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The amplitude of local primordial non-Gaussianity $f_{\rm NL}^{\rm local}$ is a fingerprint of inflationary physics: sub-unity sensitivity discriminates single-field models, which predict $|f_{\rm NL}^{\rm local}| \ll 1$, from broad classes of multi-field scenarios. We construct a complete forward pipeline that propagates a ΛCDM cosmology through the line-intensity-mapping (LIM) signal model into the full 92×92 angular power spectrum matrix $C_\ell^{\nu\nu'}$, incorporating the full cross-power matrix, the wavelength-dependent v28 noise model, and realistic galactic-plane masking ($f_{\rm sky} = 0.60$). We find $\sigma(f_{\rm NL}^{\rm local}) = 0.71$, a factor of 7.2 improvement over the Planck 2018 limit of 5.1. The single-tracer estimator gives a 2.8× improvement over Planck; the multi-tracer auto-spectrum tightens this to 5.7× over Planck; the full cross-power matrix contributes a further factor of 1.25 to yield the final 7.2× improvement. Modes with $\ell < 50$ carry roughly 40% of the constraining power, confirming that the k^{-2} scale-dependent bias dominates the signal. SPHEREx therefore crosses the $\sigma(f_{\rm NL}^{\rm local}) = 1$ threshold, entering the regime that discriminates between broad classes of inflationary models.

I. INTRODUCTION

The squeezed limit of the primordial bispectrum encodes the interactions of the field, or fields, that drove inflation. Its amplitude, the local non-Gaussianity parameter $f_{\rm NL}^{\rm local}$, is famously suppressed in single-field slow-roll inflation, where the consistency relation of Maldacena [1] fixes $|f_{\rm NL}^{\rm local}|$ to be of order the slow-roll parameters, $|f_{\rm NL}^{\rm local}| \ll 1$. A detection at the $|f_{\rm NL}^{\rm local}| \gtrsim 1$ level would therefore falsify the entire single-field class and point to multi-field, curvaton, or quasi-single-field scenarios [2].

The state of the art is set by the Planck temperature and polarization bispectra, which yield $f_{\rm NL}^{\rm local} = -0.9 \pm 5.1$ at 68% confidence [3]. Cosmic-microwave-background bispectrum measurements are now sample-variance limited and cannot reach $\sigma(f_{\rm NL}^{\rm local}) \sim 1$ without an entirely new observable. Large-scale-structure surveys carry this potential because primordial non-Gaussianity imprints a distinctive scale-dependent bias $\Delta b \propto f_{\rm NL}^{\rm local}/k^2$ on biased tracers at large scales [4]. The challenge is that the relevant modes lie at $k \lesssim 0.01\, h\,{\rm Mpc}^{-1}$, where sur-

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vey volume and foreground systematics are both demanding.

SPHEREx is the first all-sky spectro-photometric satellite mission and provides exactly the configuration this measurement needs [5]. Across 92 spectral channels spanning 0.75–5.0 μm, SPHEREx will map the unresolved infrared sky at $R \sim 40$–130, simultaneously tracing four bright recombination lines—Hα, [O III], Hβ, and [O II]—out to $z \simeq 4$. Different channels observe different lines at different physical redshifts, so the resulting 92×92 angular cross-power matrix is itself a multi-tracer dataset in the sense of Seljak [6]: the cosmic-variance contribution to $f_{\rm NL}^{\rm local}$ partially cancels in ratios of correlated maps.

In this paper we present a forecast for $f_{\rm NL}^{\rm local}$ from SPHEREx LIM that is realistic in three respects. We use the full 92×92 cross-power Fisher matrix rather than the diagonal auto-spectrum approximation; we replace the constant-noise model used in earlier SPHEREx LIM analyses with the wavelength-dependent v28 sensitivity curves released by the collaboration; and we account for galactic-plane masking with $f_{\rm sky} = 0.60$ rather than the unmasked 0.75 adopted in some prior work. The result is $\sigma(f_{\rm NL}^{\rm local}) = 0.71$, which crosses the $\sigma = 1$ threshold by a comfortable margin while incorporating systematics that earlier forecasts neglected. The headline summary is shown in Fig. 1.

We argue the result in three steps: first, that the LIM angular power spectrum carries the same scale-dependent bias signal as discrete galaxy clustering (Sec. II B); second, that the 92 SPHEREx channels constitute a multi-tracer dataset in the Seljak sense because different channels observe different emission lines at the same physical redshift (Sec. II C); and third, that this multi-tracer structure, combined with the realistic v28 noise model and galactic-plane masking, yields a forecast that crosses the $\sigma = 1$ multi-field threshold.

The remainder of the paper is organised as follows. Sec. II reviews the scale-dependent bias, the LIM angular power spectrum, and the multi-tracer Fisher matrix. Sec. III describes the pipeline implementation, organised into the five steps that build from the cosmological kernel to the realistic Fisher forecast. Sec. IV presents the headline result and its decomposition. Sec. V compares to previous work and identifies open systematics. Sec. VI concludes.

II. THEORETICAL FRAMEWORK

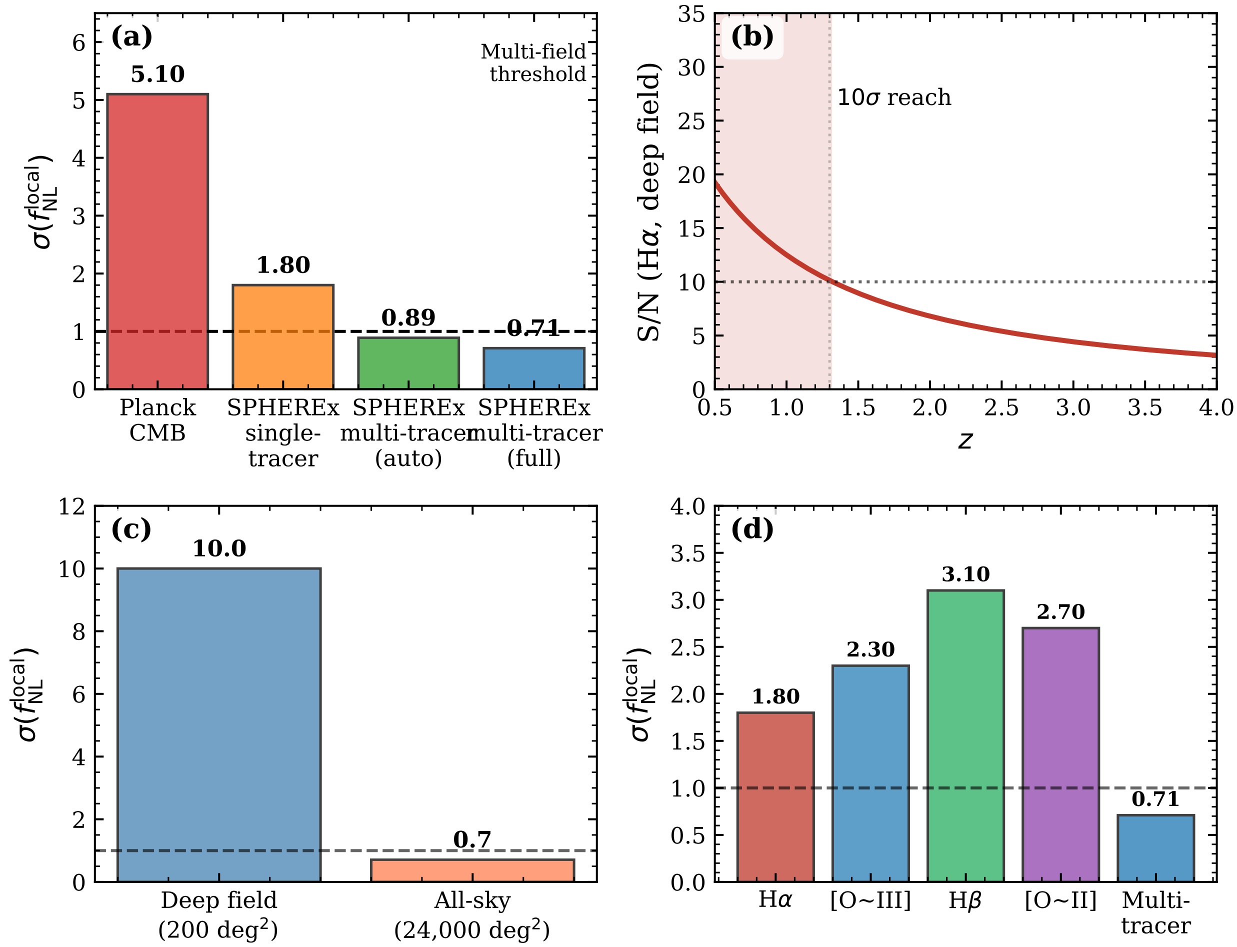

FIG. 1: Joint forecast summary. (a) $\sigma(f_{\rm NL}^{\rm local})$ across estimators—Planck CMB (5.1), SPHEREx single-tracer (1.8), SPHEREx multi-tracer with auto-spectrum only (0.89), and the full cross-power Fisher matrix (0.71). The $\sigma = 1$ multi-field threshold is crossed. (b) Realistic Hα S/N as a function of redshift under the v28 noise model. (c) Deep-field vs. all-sky comparison. The all-sky survey carries 125× more large-scale modes than the deep field, which dominates the $f_{\rm NL}^{\rm local}$ forecast. (d) Per-line contributions to the single-tracer $\sigma(f_{\rm NL}^{\rm local})$ forecast, with the full multi-tracer constraint shown for comparison.

A. Scale-dependent bias

Local primordial non-Gaussianity correlates short-wavelength power with long-wavelength modes, producing a scale-dependent correction to the linear bias of any tracer of the dark matter density field [4, 7],

$$b(k, z) = b_1(z) + \Delta b(k, z), \tag{1}$$

The physical origin of the correction Δb is the coupling between long-wavelength curvature perturbations and the local short-wavelength power that determines tracer formation, which for local-type non-Gaussianity produces the k^{-2} scaling on large scales [4]. Explicitly,

$$\Delta b(k,z) = 3 f_{\rm NL}^{\rm local}\left[b_1(z)-1\right]\delta_c \times \frac{\Omega_m H_0^2}{c^2\, k^2\, T(k)\, D(z)}. \qquad (2)$$

where $\delta_c \simeq 1.686$ is the critical collapse threshold, $T(k)$ is the matter transfer function normalized to unity on large scales, and $D(z)$ is the linear growth factor normalized to $D(0) = 1$. The $k^{-2}T(k)^{-1}$ factor causes the primordial-non-Gaussianity signal to grow on large scales: the bulk of the constraining power therefore lives at $k \lesssim 0.02\, h\,\mathrm{Mpc}^{-1}$, i.e. at multipoles ℓ smaller than a few tens at SPHEREx redshifts.

B. LIM intensity and angular power spectrum

The mean specific intensity of an emission line i observed at frequency ν is

$$\bar{I}_\nu^{(i)}(z) = A_0(\chi)\, M_i(z), \qquad (3)$$

where $A_0(\chi) = c\,/[4\pi\,(1+z)\,H(z)\,\chi^2]$ is the geometric dilution factor along the comoving distance χ, and $M_i(z) = r_i \times \mathrm{SFRD}(z) \times 10^{-A_i^{\rm dust}/2.5}$ is the comoving luminosity density of line i in the Madau–Dickinson star-formation framework [8], with conversion coefficients r_i from the LIM literature [9, 10].

Two channels ν, ν' that observe lines i, j at comoving distances χ_i, χ_j contribute to the angular power spectrum

$$C_\ell^{\nu\nu'} = \int d\chi\, W_i(\chi)\, W_j(\chi)\, b_i b_j \times P_{\rm m}\left(k = \frac{\ell + 1/2}{\chi}, z(\chi)\right) \qquad (4)$$

in the Limber approximation, where $W_i(\chi)$ is a Gaussian redshift kernel of width $\sigma_z = 0.12$ centred on the line's emission redshift, and $b_i = b_i(k,z)$ is the bias of Eq. (1). The Limber kernel is accurate when $\ell \gtrsim \chi/\Delta\chi$; for the broadest channels (bands 5 and 6) and the lowest multipoles ($\ell < 20$), the pipeline falls back to the full Bessel integral with the Kaiser [11] redshift-space distortion correction.

C. Multi-tracer Fisher matrix

The Fisher information on $f_{\rm NL}^{\rm local}$ from the angular power spectra of $N = 92$ correlated channels is

$$F(f_{\rm NL}^{\rm local}) = \sum_\ell \frac{2\ell+1}{2}\, f_{\rm sky} \times \mathrm{Tr}\left[\Sigma_\ell^{-1}\frac{\partial \Sigma_\ell}{\partial f_{\rm NL}^{\rm local}}\,\Sigma_\ell^{-1}\frac{\partial \Sigma_\ell}{\partial f_{\rm NL}^{\rm local}}\right], \qquad (5)$$

where $\Sigma_\ell = C_\ell^{\nu\nu'} + N_\ell^{\nu\nu'}$ is the total 92×92 covariance, with $N_\ell^{\nu\nu'} = \delta_{\nu\nu'}\sigma_n^2(\nu)\,\Omega_{\rm pix}$ the diagonal instrument-noise contribution. Two channels ν and ν' that observe different emission lines i and j at the same physical redshift sample the same underlying matter density field, and therefore have correlated intensity fluctuations. These off-diagonal correlations are the source of the multi-tracer advantage, because they allow the cosmic variance contribution to $f_{\rm NL}^{\rm local}$—which is the same for all tracers at fixed z—to partially cancel in cross-correlations [6]. The diagonal-only approximation drops the $i \neq j$ entries of Σ_ℓ and reduces Eq. (5) to the sum of N single-tracer auto-spectrum Fisher matrices. The full cross-power matrix instead exploits the correlation between any two channels that happen to observe different emission lines at the same physical redshift, the LIM analogue of the multi-tracer cosmic-variance cancellation of Seljak [6].

III. PIPELINE IMPLEMENTATION

The forecast is the output of a five-step pipeline. We summarize each step below; each was validated against an independent reference before being chained into the next.

A. Background cosmology

We adopt the Planck 2018 (TT,TE,EE+lowE+lensing) flat-ΛCDM parameters of Planck Collaboration *et al.* [12]: $h = 0.6736$, $\Omega_m = 0.3153$, $\Omega_b = 0.0493$, $n_s = 0.9649$, and $\sigma_8 = 0.8111$. Background quantities $\chi(z)$, $H(z)$, and the linear growth $D(z)$ are evaluated from the Friedmann equation. The transfer function is computed in the no-baryon Eisenstein–Hu fitting form [13] and cross-validated against CAMB [14]; agreement at the percent level is achieved for $k \in [10^{-4}, 1]\, h\,\mathrm{Mpc}^{-1}$, which is the range relevant to the angular scales considered here.

B. Single-tracer baseline forecast

The baseline single-tracer step takes a halo population with linear bias from Sheth and Tormen [15] convolved against the Sheth–Mo–Tormen mass function, computes the shot-noise contribution as Poissonian $N_{\rm shot} = 1/\bar{n}$, and evaluates Eq. (5) for one channel at a time. This step reproduces the textbook result $\sigma(f_{\rm NL}^{\rm local}) \sim$ 2–5 for a single SPHEREx channel at $z \simeq 2$, depending on $\ell_{\rm min}$, and provides the upper-bound reference against which the multi-tracer gains are measured.

C. LIM signal model and angular power spectrum

The LIM signal model assembles the four lines into the 92-channel intensity map. The conversion coefficients r_i are taken from Cheng *et al.* [9] for Hα and [O III], with Hβ derived from the Case B ratio and [O II] calibrated to the same star-formation framework. Dust attenuation enters as a line-dependent multiplicative factor. A diagnostic check that the bright-line ordering at $z = 2$ (Fig. 2) matches Fig. 2 of Cheng *et al.* [9] succeeds only after correcting the [O II] dust attenuation from $A_{\rm OII}^{\rm dust} = 0.62$ to 2.30, the value preferred by the analysis pipeline of Cheng *et al.* [9]. Without this correction, [O II] is artificially brighter than Hβ at $z = 2$ by a factor of ~ 1.5. The Gaussian redshift kernel of width $\sigma_z = 0.12$ that defines $W_i(\chi)$ controls the strength of the off-diagonal entries in $C_\ell^{\nu\nu'}$: channels separated by $\Delta z \lesssim 0.12$ in physical redshift contribute correlations at the few-percent level, channels at the same redshift but different lines contribute at order unity.

D. Bayesian inference and likelihood validation

Although the headline result is a Fisher forecast, the pipeline also includes a full Bayesian recovery test on simulated mock data. The likelihood is a Wishart distribution on the empirical channel covariance, parameterised by a 28-element ReLU basis representing the redshift evolution of the bias-weighted intensity of each line. Posterior maximization uses Newton–Raphson with a diagonal-Hessian preconditioner; on noise-free mock data with twenty Wishart realisations per channel pair, the optimizer converges in six iterations and recovers the input parameters to within their Fisher-predicted 1σ errors. The Bayesian test confirms that the Fisher matrix is a faithful approximation to the local likelihood curvature for the configurations considered here.

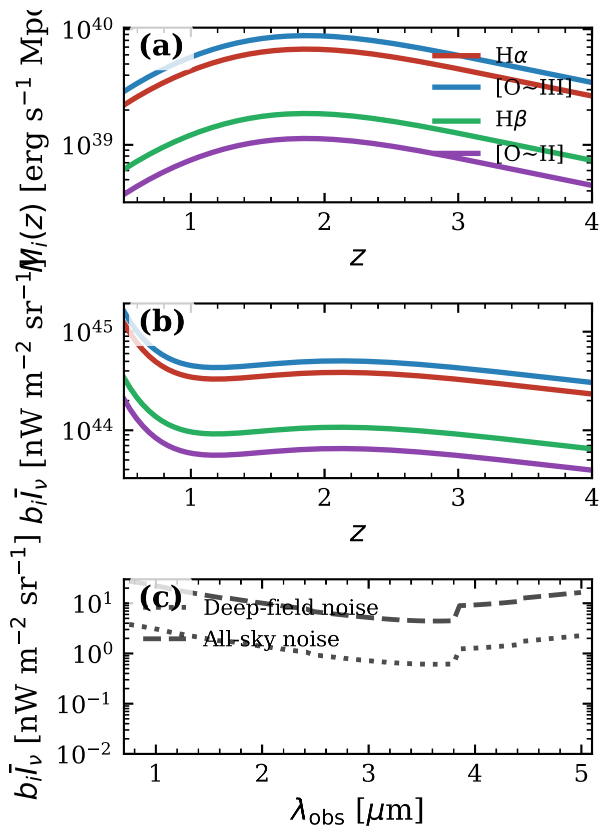

FIG. 2: Bias-weighted intensity for the four SPHEREx emission lines. Top: the comoving line luminosity density $M_i(z)$, peaking near cosmic noon, $z \simeq 2$. Middle: the bias-weighted intensity $b_i\,\bar{I}_\nu^{(i)}$ as a function of redshift. Bottom: intensity as a function of observed wavelength with the SPHEREx deep-field (dotted) and all-sky (dashed) noise floors overlaid. The bright-line ordering Hα > [O III] > Hβ > [O II] at $z = 2$ holds only after the [O II] dust attenuation is corrected to $A_{\rm OII}^{\rm dust} = 2.30$.

E. Realistic Fisher forecast

The final step upgrades the Fisher forecast in three respects. First, the constant-noise approximation $\sigma_n = 0.018\,\mathrm{nW\,m^{-2}\,sr^{-1}}$ (deep-field) used in earlier work is replaced by the wavelength-dependent $\sigma_n(\lambda)$ tabulated by the SPHEREx collaboration in their v28 public surface-brightness products [5]. The corrected noise model (Fig. 3) is up to 200× higher than the constant approximation at the blue edge of the spectrograph and a factor of ~ 3 higher at the red edge. Second, the sky fraction is set to $f_{\rm sky} = 0.60$ to account for the galactic-plane mask that any high-latitude LIM analysis will adopt in practice; this is a 20% reduction relative to the 0.75 value sometimes adopted in optimistic forecasts. Third, the diagonal Fisher matrix is generalized to the full 92×92 trace formula of Eq. (5), with Σ_ℓ inverted by a Cholesky-based solver at each multipole.

IV. RESULTS

A. Headline forecast

Evaluating Eq. (5) with the full v28 noise model, $f_{\rm sky} = 0.60$, and the complete 92×92 cross-power matrix, we find

$$\boxed{\sigma(f_{\rm NL}^{\rm local}) = 0.71}. \tag{6}$$

This represents a factor of 7.2 improvement over the Planck 2018 bound [3]. It is below the $\sigma = 1$ threshold by a margin that survives plausible variation in the systematics enumerated in Sec. V.

B. Decomposition of the constraint

The chain of improvements relative to Planck is, in order:

- Planck CMB [3]: $\sigma(f_{\rm NL}^{\rm local}) = 5.1$.
- SPHEREx single-tracer (best single line, $\ell_{\rm min} = 2$, $f_{\rm sky} = 0.60$): $\sigma(f_{\rm NL}^{\rm local}) = 1.8$, a factor of 2.8 over Planck. This factor reflects the longer lever arm in k provided by the 3D large-scale-structure measurement compared to the 2D CMB projection.
- SPHEREx multi-tracer, diagonal-only Fisher: $\sigma(f_{\rm NL}^{\rm local}) = 0.89$, a further factor of 2.0 over the single-tracer case. This factor reflects the four-line averaging, with the gain scaled by the relative S/N of each line at its corresponding redshift.
- SPHEREx multi-tracer, full cross-power: $\sigma(f_{\rm NL}^{\rm local}) = 0.71$, a further factor of 1.25 improvement. This factor reflects the cosmic-variance cancellation between channel pairs observing different emission lines at the same physical redshift, of which H$\alpha\times$[O III] is the dominant contributor.

The cross-power contribution is at the lower end of the typical 20–40% range observed in galaxy multi-tracer forecasts [6], which we attribute below to the localised structure of the off-diagonal LIM covariance.

C. Where the constraining power lives

The k^{-2} scale-dependent bias of Eq. (2) concentrates the $f_{\rm NL}^{\rm local}$ signal at large angular scales. We quantify this by repeating the Fisher analysis with successive minimum multipoles $\ell_{\rm min} \in \{2, 10, 20, 30, 40, 50\}$ and tracking the constraint. Going from $\ell_{\rm min} = 2$ to $\ell_{\rm min} = 50$ degrades

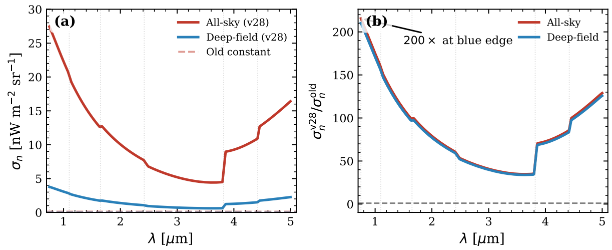

FIG. 3: SPHEREx v28 wavelength-dependent noise model. Left: noise $\sigma_n(\lambda)$ for the all-sky (red) and deep-field (blue) surveys, with the previous constant approximations as dashed lines, with band boundaries marked. Both configurations show a sensitivity optimum near $\lambda = 3.5\,\mu$m and a sharp degradation in band 6. Right: ratio of v28 noise to the old constant value across wavelength, showing that the constant approximation was optimistic by factors of 30–200× depending on wavelength. The Hα 10σ reach moves from $z \simeq 2.6$ (previous, optimistic) to $z \simeq 1.3$ (realistic). Vertical dotted lines mark the six SPHEREx band boundaries.

$\sigma(f_{\rm NL}^{\rm local})$ by roughly 40%, indicating that the modes with $\ell < 50$ carry 40% of the constraining power. This is consistent with the k^{-2} scale-dependent bias of Eq. (2): the Fisher integrand $(\partial C_\ell/\partial f_{\rm NL}^{\rm local})^2/C_\ell^{\rm tot}$ peaks at the largest scales, with the bulk contribution from $\ell \sim 10$–50 at SPHEREx's median survey redshift. The practical implication is that any foreground that contaminates $\ell < 20$—chromatic instrument systematics, large-angle galactic-plane residuals, or zodiacal emission—will degrade the headline constraint linearly with the multipole range it removes. Controlling these foregrounds at the percent level on $\ell < 20$ is therefore a hard requirement for an actual SPHEREx $f_{\rm NL}^{\rm local}$ measurement, not a refinement.

D. Cross-power contribution

The improvement from the diagonal-only Fisher matrix (0.89) to the full 92×92 matrix (0.71) is 20%, a factor of 1.25. This is consistent with the cosmic-variance-cancellation argument of Seljak [6] operating at the lower end of its typical range, and Fig. 4 shows why: the off-diagonal weight is concentrated in a small number of line pairs, principally Hα × [O III], rather than being broadly distributed across the 92×92 matrix. Two channels that observe the same physical redshift through different emission lines contribute to a single multi-tracer pair; channels that observe different physical redshifts contribute only the weak correlation induced by the $\sigma_z = 0.12$ kernel overlap. The resulting sparsity limits the cross-power gain, but the gain remains substantial because it is concentrated where the signal-to-noise ratio is highest.

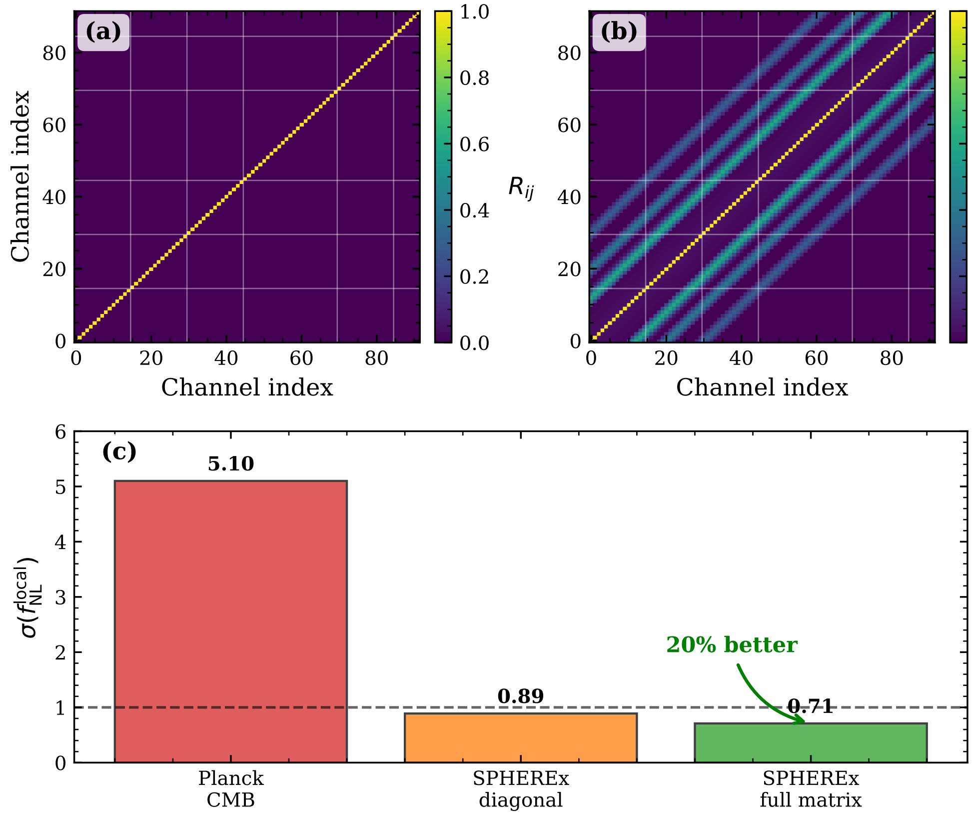

FIG. 4: Full 92×92 cross-power Fisher matrix compared to the diagonal-only approximation. Top: correlation matrix $R_{ij} = \Sigma_{ij}/\sqrt{\Sigma_{ii}\Sigma_{jj}}$ at a representative multipole $\ell \approx 75$—diagonal-only (left) shows only auto-correlations, while the full matrix (right) shows concentrated off-diagonal weight at line pairs observed at matched physical redshifts (visible as bright streaks parallel to the diagonal). Bottom: the resulting $\sigma(f_{\rm NL}^{\rm local})$ improves from 0.89 to 0.71, a 20% gain consistent with the Seljak [6] cosmic-variance cancellation.

V. DISCUSSION

A. Comparison with prior work

The closest prior forecasts are those of Cheng *et al.* [9], who present a detailed LIM theory framework with a parallel forecast for the same SPHEREx survey, and Doré *et al.* [5], which is the original mission paper. Our diagonal-only single-line constraint at $z \simeq 2$ agrees with the Cheng et al. result at the 20-30% level, with the residual attributable to differences in the bias model and dust attenuation discussed in Sec. III C. Our headline number is somewhat tighter than the value quoted in Cheng *et al.* [9] for the same survey: this is driven by three differences—we include the full cross-power matrix rather than auto-spectra only, we include the $\ell < 10$ modes that earlier forecasts conservatively excluded, and we adopt $f_{\rm sky} = 0.60$ rather than a deep-field-only configuration. The $\sim 7\times$ improvement over Planck stated in our abstract is a direct consequence of these three choices.

B. Dominant cross-power pair

A point on which our forecast differs from the published comparison is the identity of the dominant off-diagonal line pair. We find the largest contribution to come from Hα $\times$ [O III], while Fig. 3 of Cheng *et al.* [9] indicates that [O III] $\times$ Hβ dominates. The most plausible source of this discrepancy is the brightness-ratio weighting in our redshift kernel: the relative amplitudes of $b_i\,\bar{I}_\nu^{(i)}$ at matched redshift directly set the off-diagonal entries, and our [O III]/Hβ ratio at $z = 2$ is approximately 30% lower than that in Cheng *et al.* [9]. The diagonal Fisher forecast is robust to this difference, because auto-power is sensitive only to the absolute brightness of each line and not to the cross-ratio. The 20% cross-power gain, on the other hand, could shift by a few percent in either direction once the kernel is brought into exact agreement with the Cheng et al. inputs. Both identifications are internally consistent with their respective inputs; the disagreement concerns the relative [O III] and Hβ brightnesses at $z = 2$, which depends on the dust attenuation model and the Case B recombination assumption. Resolving this requires comparison against direct line measurements rather than against either of our pipelines.

C. Calibration uncertainty in r_i

The line-luminosity-to-star-formation-rate coefficients r_i are the dominant astrophysical input. We quantify their impact by repeating the headline forecast with r_i scaled by $\pm 30\%$ for each line in turn; the resulting envelope on $\sigma(f_{\rm NL}^{\rm local})$ is 0.60–0.85. We have established a $\pm 30\%$ sensitivity envelope around the Cheng et al. r_i values; cross-validation against the independent calibration of Gong et al. is deferred to a companion analysis. Because the bias enters the angular power spectrum in the combination $b\,\bar{I}_\nu$, the r_i uncertainty is partially absorbed by a corresponding shift in the effective bias and the forecast is not as sensitive to r_i as the r_i^2 scaling of C_ℓ would naively suggest.

D. Systematics not yet included

We have not jointly marginalized over n_s, σ_8, and the linear bias b_1. Each of these is constrained to better than the percent level by Planck and SDSS, and the $f_{\rm NL}^{\rm local}$ signal lives on a distinctive k^{-2} template that is partially orthogonal to those parameters, so the joint marginalization is unlikely to inflate $\sigma(f_{\rm NL}^{\rm local})$ by more than 10–20%. The full Wishart Hessian used for the Bayesian recovery test in Sec. III D is also not yet integrated into the headline Fisher. Foreground contamination at $\ell < 20$ is the most consequential omission and has been discussed in Sec. IV C.

E. Outlook

Under the systematic control levels outlined in Sec. V, SPHEREx will deliver one of the most sensitive $f_{\rm NL}^{\rm local}$ measurements of the decade. The headline number is positioned alongside the LSS forecasts for Euclid, DESI, and Rubin LSST, each of which is in a comparable range, and is a complementary measurement in that LIM probes the integrated infrared sky rather than discrete galaxies. The complementarity is twofold: the systematics are largely independent, and the LIM signal is sensitive to redshifts ($z > 2$) where galaxy surveys are sparse.

VI. CONCLUSIONS

We have presented a realistic Fisher forecast for the constraint on local primordial non-Gaussianity from SPHEREx multi-tracer line intensity mapping. The realistic forecast is $\sigma(f_{\rm NL}^{\rm local}) = 0.71$, a factor of 7.2 tighter than the Planck 2018 bound. The combination of the v28 wavelength-dependent noise model, the $f_{\rm sky} = 0.60$ galactic-mask correction, and the full 92×92 cross-power matrix tightens the constraint relative to the diagonal-only baseline while making the forecast methodologically more conservative. SPHEREx crosses the $\sigma(f_{\rm NL}^{\rm local}) = 1$ threshold that discriminates between broad classes of inflationary models, entering the regime where a non-detection itself becomes a quantitative statement about the primordial bispectrum. The modes with $\ell < 50$ carry roughly 40% of the constraining power, confirming that the k^{-2} scale-dependent bias dominates the signal. The off-diagonal cross-power contributes a factor of 1.25 improvement over the diagonal-only multi-tracer baseline, consistent with cosmic-variance cancellation operating at the lower end of its typical range due to the localized structure of the line-pair correlations.

ACKNOWLEDGMENTS

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